Open problems:
Schrödinger equation is equivalent to information conservation

Open problems raised by the technical brief in this directory, and by its Lean formalization (see `formalization-log.tex`).

From the brief

1. **Classical counterpart.** State and prove the classical analogue: Hamiltonian evolution is equivalent to preservation of the Shannon entropy and of "the definitional independence of degrees of freedom". (Brief, Sec. 4: "This parallels the classical case where Hamiltonian evolution is equivalent to the preservation of Shannon entropy (Liouville's theorem)".)

2. **Entropy conservation as determinism and reversibility.** The brief reads DR-INFO as "an information theoretic way to characterize determinism and reversibility". A precise definition of deterministic and reversible evolution, and a proof that it is equivalent to DR-INFO, are not given. (Brief, Sec. 1.)

3. **Propagator for unbounded time-dependent $H(t)$.** "The Schrödinger equation gives rise to a differentiable unitary propagator $U_{t_0 \to t_1}$" is assumed. For a continuous family of bounded self-adjoint $H(t)$ this is now proved, in any dimension, from the global existence and uniqueness theorem for linear ODEs: the propagator exists, is unique and is unitary (`exists_propagator_of_continuous`, `propagator_unique`), and in finite dimension its two-parameter evolution satisfies [B]EVMIX, DR-SCEQ, DR-INFO and EVDIFF (`TwoParameter.exists_propagator_isEvolution`; log, "The propagator of a time-dependent Hamiltonian"). Existence, uniqueness and unitarity also hold for bounded $H(t)$ that are only strongly continuous, with $t \mapsto U(t)\psi$ differentiable for every ψ (`exists_propagator_of_continuous_apply`), and so does EVDIFF: $U(t)\rho U(t)^\dagger$ is differentiable in operator norm for every density operator ρ (`hasDerivAt_conj_of_continuous_apply`). The propagator itself need not be norm-differentiable: on $L^2[0,1]$, multiplication by the indicator of $[0,t]$ is such an $H(t)$, and no propagator of it is norm-differentiable on $(0,1)$ (`indicatorHamiltonian_counterexample`). For unbounded self-adjoint $H(t)$ existence needs Kato-type hypotheses and remains open. The converse has the same gap: for unbounded $H(t)$, Step 3 gives only a symmetric $H(t_0)$ (next section, first item). (Brief, Sec. 3, proof of Theorem 2 ($\Rightarrow$); Log, "Differences from the brief".)

From the formalization

Beyond finite dimension the formalization has a part in any dimension; it leaves open the items "Unbounded time-dependent Hamiltonians", "Equation at every time in infinite dimension" and "Wold–Cooper reduction" below. Whether a phase choice makes $t \mapsto U(t)$ differentiable in operator

norm in infinite dimension is settled negatively: on $\ell^2(\mathbb{N})$, a phase e^{it} on the basis vector $e_{\lceil 1/|t| \rceil + 1}$ satisfies all hypotheses, with EVDIFF in operator norm, but no unitary family inducing it is norm-differentiable at 0 (`movingPhase_counterexample`; log, "No norm-differentiable propagator").

1. **Unbounded time-dependent Hamiltonians.** EVDIFF on every density operator forces a bounded H (Banach–Steinhaus). With EVDIFF on a dense set of states, Theorem 2 is formalized for a time-independent evolution, with unbounded self-adjoint H (`drSceq_iff_drInfo_and_evDiff`, by Stone's theorem). Without EVTI, Step 3 gives only a skew-symmetric $A(t_0)$, and $H(t_0)$ need not be self-adjoint: the propagator of $H(t) = H_\beta$ for $t < 0$ and H_α for $t \geq 0$, two self-adjoint extensions of the momentum on $L^2(0,1)$, is differentiable at 0 exactly on the domain of the minimal momentum operator, which is not self-adjoint. Open: the hypotheses (Kato-type, or on the evolution) under which DR-INFO + EVDIFF gives a self-adjoint $H(t)$. The example is formalized: the vectors on which the propagator is differentiable at 0 form a dense subspace, on which $i\hbar U'(0)$ is symmetric but not self-adjoint (`switchedHamiltonian_counterexample`). This subspace is $H_0^1(0,1)$, the domain of the minimal momentum, and $i\hbar U'(0)$ is $-i\,d/dx$ on it (`switchedDomain_eq`). The two pieces of U are $e^{-itH_\alpha/\hbar}$ and $e^{-itH_\beta/\hbar}$ (`twistedTranslation_eq_schrodingerPropagator`). (Log, "Unbounded Hamiltonians".)

   The example uses the two-sided derivative at $t_0 = 0$, and it splits the problem in two. Each one-sided derivative alone has a self-adjoint generator. The right derivative of $e^{-itH_\alpha/\hbar}\psi$ at 0 exists exactly for ψ in the domain of H_α, and the left derivative of $e^{-itH_\beta/\hbar}\psi$ exactly for ψ in the domain of H_β (`exists_hasDerivWithinAt_iff`). The domain of the two-sided generator is the intersection of the two, so it is too small; it is a common restriction of H_α and H_β (`switchedHamiltonian_le_momentum_left`, `switchedHamiltonian_le_momentum_right`).

   (a) *Two-sided derivative*, as in Step 3: DR-INFO + EVDIFF do not give a self-adjoint $H(t_0)$. This is settled by the example above.

   (b) *Forward derivative only.* The evolution $\Phi_{t_0 \to t_1}$ is only given for $t_0 \leq t_1$, so one can define $H(t_0)$ by the right derivative of $t \mapsto \Phi_{t_0 \to t}$ at t_0. The example above says nothing here: its forward generator at 0 is H_α, which is self-adjoint. A continuously varying boundary phase settles it negatively (`sqrtPhase_counterexample`). Take $H(t) = H_{\alpha(t)}$ with $\alpha(t) = \alpha_0 + \sqrt{t}$. Its propagator is unitary and satisfies $U(t_2,t_1)U(t_1,t_0) = U(t_2,t_0)$, and at every $t_0 \neq 0$ the forward generator is the self-adjoint $H_{\alpha(t_0)}$. At $t_0 = 0$ the forward domain is the dense $H_0^1(0,1)$, and the forward generator is the minimal momentum, symmetric but not self-adjoint (`forwardDomain_sqrtPhase`, `not_isSelfAdjoint_forwardHamiltonian_sqrtPhase`). Hölder exponent $\frac{1}{2}$ is the threshold: if $|\alpha(t_0+s) - \alpha(t_0)| = o(\sqrt{s})$, the forward generator is $H_{\alpha(t_0)}$ (`forwardHamiltonian_eq_momentum`). So what is missing is regularity of the evolution in time. At a single time a hypothesis on the evolution alone is formalized. The forward generator of any unitary evolution is symmetric, and it is self-adjoint iff at t_0 the evolution is tangent to a one-parameter unitary group: $U(t_0+\tau,t_0)\psi - e^{-i\tau H/\hbar}\psi = o(\tau)$ for a self-adjoint H and every ψ in its domain, and then $H = H(t_0)$ (`isSelfAdjoint_forwardHamiltonian_iff`). In the example no group is tangent at 0 (`not_exists_tangent_sqrtPhase`). A condition on the evolution alone that gives an essentially self-adjoint forward generator is formalized: a dense subspace D invariant under all $U(t,t_0)$, on which the orbits are differentiable and the generators near t_0 differ from the one at t_0 by operators bounded on D (`isEssentiallySelfAdjoint_forwardHamiltonian`). If the differences also tend to 0 as $t \to t_0^+$, the forward generator itself is self-adjoint, by a Duhamel estimate

(`isSelfAdjoint_forwardHamiltonian`). The example has no such D. Open: a relative bound in place of a bounded difference, as for a time-dependent unbounded potential. The usual meaning of "U solves the Cauchy problem" is not enough: dense subspaces Y_t with $U(t,s)Y_s \subseteq Y_t$ on which $t \mapsto U(t,s)\psi$ is C^1 (Engel–Nagel, Def. VI.9.2). In the example $Y_0 = H_0^1(0,1)$ and $Y_t = D(H_{\alpha(t)})$ for $t > 0$ are such subspaces, by direct computation (not formalized): for $\psi \in H_0^1$, $|\psi(1-u)| = o(\sqrt{u})$ compensates the derivative $\frac{1}{2\sqrt{u}}$ of the phase. Engel and Nagel note that no analogue of the generation theorem is known for time-dependent problems (Sec. VI.9.a). Kato's theorem gives the $\Leftarrow$ direction under its hypotheses (constant domain, $t \mapsto H(t)\psi$ of class C^1); in the example the domain is not constant.

What holds without hypotheses on $H(t)$ is Howland's construction: a strongly continuous unitary propagator $U(t,s)$ gives the strongly continuous unitary group $(W(\sigma)f)(t) = U(t, t-\sigma)f(t-\sigma)$ on $L^2(\mathbb{R}; E)$, hence a self-adjoint quasi-energy operator K with $W(\sigma) = e^{-i\sigma K/\hbar}$ (`UnitaryPropagator.exists_howlandGenerator`). On compactly supported f whose moving orbit $U(t, t-\sigma)f(t-\sigma)$ is C^1 in σ, Kf is $i\hbar$ times its σ-derivative at 0 (`UnitaryPropagator.howlandGenerator_toLp`). For a time-independent evolution this gives $K = H - i\hbar\,\partial_t$ on χx (`UnitaryPropagator.howlandGenerator_smul_of_group`). Open: how K determines $H(t_0)$ at a single time, and whether that $H(t_0)$ is self-adjoint.

2. **Equation at every time in infinite dimension.** In finite dimension, one differentiable unitary $U(t)$ for all times solves the Schrödinger equation at every time, and $\Phi_{t_0 \to t_1}$ is conjugation by $U(t_1)U(t_0)^\dagger$ (`TwoParameter.drSceq_iff_drInfo_and_evDiff`). The proof fixes the determinant of the local unitaries, which leaves only a root of unity to match when gluing them. In any dimension the determinant is not defined, and the local unitaries are only strongly differentiable, and in general cannot be chosen norm-differentiable (`movingPhase_counterexample`), so the global propagator is not built there. (Log, "The equation at every time".)

3. **Wold–Cooper reduction.** The right shifts V_t on $L^2(0,\infty)$ give an evolution that satisfies [B]EVMIX, maps pure states to pure states, preserves transition probabilities and the entropy, but is induced by no unitary; this is formalized (`shift_semigroup_counterexample`, in the brief "Multiple equivalent characterizations of the Schrödinger equation"). The Wold–Cooper decomposition, which reduces every strongly continuous semigroup of isometries to a unitary group plus copies of this one, is not formalized. (Log, any-dimension part, "Problems found in the brief", second item.)